A MULTISYMPLECTIC APPROACH TO BOUNDARY TERMS

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Introduction and motivations

When computing conserved charges of field theories, boundary terms may arise when applying Noether's theorem if field configurations do not decay fast enough at infinity. Some conserved quantities, such as *electric charge* or *electromagnetic and gravitational energy and momentum*, have a non-trivial **boundary term**, while others such as *global*$-U(1)$ *charge* or *scalar field energy and momentum* do not.

In opposition to symplectic formalism of field theory, in which fields are interpreted as an infinite set of degrees of freedom evolving in time, in **multisymplectic formalism** fields are understood as an unified degree of freedom that extend throughout the spacetime. Because of the local nature of multisymplectic formalism, it is natural to ask how boundary terms in the conserved charges arise in this formulation.

Multisymplectic formalism

Let the dynamics of a system of fields $\{y^i\}$ over a $n-$dimensional manifold $\mathcal{M}$ with coordinates x^μ be described by a least action principle with lagrangian density $\mathcal{L}(y^i, \partial_\mu y^i, x^\mu)$. By performing a Legendre transform with respect to all the manifold coordinates $\mathcal{H} \equiv p_i^\mu \partial_\mu y^i - \mathcal{L}$, where $p_i^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu y^i)}$ are the multimomenta, we obtain the De Donder-Weyl-Hamilton equations[1]:

$$\partial_\mu y^i = \frac{\partial \mathcal{H}}{\partial p_i^\mu}, \qquad \partial_\mu p_i^\mu = -\frac{\partial \mathcal{H}}{\partial y^i}$$

The geometry of the manifold (x^μ, y^i, p_i^μ) in which the dynamics take place is characterized by a multisymplectic form Ω, whose expression in local coordinates is given by:

$$\Omega = dy^i \wedge dp_i^\mu \wedge d^{n-1}x_\mu + d\mathcal{H} \wedge d^n x$$

which has a multisymplectic potential $\Omega = -d\Theta$, with coordinates $\Theta = p_i^\mu dy^i \wedge d^{n-1}x_\mu - \mathcal{H} d^n x$. This geometric structure defines a mapping from $p-$vectors V to $(n-p)-$forms F on multisymplectic phase space via:

$$\iota_V \Omega = dF$$

Noether's theorem[2] states that if the vector field ξ is a **symmetry** of the hamiltonian up to a boundary term $d\alpha$, then the $(n-1)-$form $F_\xi = \iota_\xi \Theta - \alpha$ is a **conserved quantity** of every solution of the equations of motion $(j^1\phi)(x^\mu) = (x^\mu, y^i(x^\mu), p_i^\mu(x^\mu))$:

$$\mathcal{L}_\xi \Theta = d\alpha \Longrightarrow (j^1\phi)^* d(\iota_\xi \Theta - \alpha) = 0$$

The $(n-1)-$symmetry form F_ξ can be **integrated** on a $(n-1)-$dimensional **submanifold** Σ, resulting in a conserved charge:

$$Q_\xi = \int_\Sigma \mathrm{i}^*(j^1\phi)^* F_\xi$$

where $\mathrm{i}: \Sigma \to \mathcal{M}$ is the inclusion map.

How will boundary terms appear?

Infinitesimal transformations on the multisymplectic phase space are generated by vector fields ξ, usually proportional to a (position dependent) parameter. Some symmetries, however, involve not only this parameter, but also its **derivatives**, which could also appear in the multisymplectic conserved $(n-1)-$form F_ξ and in the conserved charge Q_ξ.

To ensure that the conserved charge Q_ξ associated with this symmetry is proportional to its parameter, one may perform **integration by parts** on the terms involving spatial derivatives of the parameter. This process shifts the derivatives onto other fields, isolating the parameter, and resulting in additional **boundary terms** to the charge.

Complex scalar field

The hamiltonian of a complex scalar field is:

$$\mathcal{H} = p_\mu^* p^\mu + m^2 \phi^* \phi$$

which gives as equations of motion the following:

$$\partial_\mu \phi = p_\mu^*, \qquad \partial_\mu p^\mu = -m^2 \phi^*$$

This hamiltonian is invariant under two transformations: global$-U(1)$ rephasing and Poincaré. Rephasing symmetry is given by the transformation $\phi \to e^{i\alpha}\phi$ and $p^\mu \to e^{-i\alpha} p^\mu$, therefore generated by the following vector field on the multisymplectic phase space:

$$V_\alpha = i\alpha\phi\partial_\phi - i\alpha p^\mu \partial_{p^\mu} - i\alpha\phi^*\partial_{\phi^*} + i\alpha p^{*\mu}\partial_{p^{*\mu}}$$

The conserved multisymplectic $(n-1)-$form is:

$$F_\alpha = i\alpha(\phi p^\mu - \phi^* p^{*\mu}) d^{n-1}x_\mu$$

and the conserved charge integrated on the constant-time slice Σ_t:

$$\boxed{Q_\alpha = \int_{\Sigma_t} i\alpha\left(\phi\pi - \phi^*\pi^*\right) d^3x}$$

where we have defined $\pi = p^0$ as the symplectic momentum, normal to the hypersurface Σ_t.

Poincaré transformations of the scalar field are generated by the following vector[3]:

$$V_\xi = -\xi^\mu \partial_\mu - (p^\nu \partial_\nu \xi^\mu - p^\mu \partial_\mu \xi^\nu)\partial_{p^\mu} - (p^{*\nu}\partial_\nu \xi^\mu - p^{*\mu}\partial_\nu \xi^\nu)\partial_{p^{*\mu}}$$

The conserved multisymplectic $(n-1)-$form is:

$$G_\xi = \xi^\mu\left(p^\nu d\phi \wedge d^{n-2}x_{\mu\nu} + p^{*\nu} d\phi^* \wedge d^{n-2}x_{\mu\nu} + \mathcal{H} d^{n-1}x_\mu\right)$$

and the conserved charge integrated on the constant-time slice Σ_t:

$$\boxed{Q_\xi = \int_{\Sigma_t} \xi^\mu T_\mu^0 d^3x}$$

where $T_\mu^\nu \equiv p^\nu \partial_\mu \phi + p^{*\nu}\partial_\mu \phi^* - \delta_\mu^\nu \mathcal{L}$ is the energy-momentum tensor.

Electromagnetism

The lagrangian of the electromagnetic field is:

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu}$$

where $F^{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The multimomenta are given by $p_{A_\nu}^\mu = -F^{\mu\nu}$, so they are antisymmetric, i.e., $p^{\mu\nu} = -p^{\nu\mu}$, so the system is degenerate. This constraints lead to the constraints $\pi^0 = 0$ and $\partial_i \pi^i = 0$ in the constant-time surface, where $\pi^\mu = p^{0\mu}$ are the symplectic momenta.

Electromagnetism is invariant under two transformations: local gauge transformation and Poincaré.

Gauge symmetry is given by the transformation $A_\mu \to A_\mu + \partial_\mu \varepsilon$, generated in phase space by:

$$V_\varepsilon = \partial_\mu \varepsilon \partial_{A_\mu}$$

The conserved multisymplectic $(n-1)-$form is:

$$F_\varepsilon = p_{A_\nu}^\mu \partial_\nu \varepsilon d^{n-1}x_\mu$$

and the conserved charge, once integrated by parts to extract the parameter out of the derivative:

$$\boxed{Q_\varepsilon = \int_{\Sigma_t} \pi^0 \partial_0 \varepsilon d^3x - \int_{\Sigma_t} \varepsilon \partial_i \pi^i d^3x + \oint_{\partial\Sigma_t} \varepsilon \pi^i d^2x_i}$$

The two first terms are zero due to the constraints, so the electric charge is just a boundary term.

Poincaré transformations of the electromagnetic field are generated by the following vector[3]:

$$V_\xi = -\xi^\mu \partial_\mu + A_\nu \partial_\mu \xi^\nu \partial_{A_\mu} - \left(p_{A_\nu}^\rho \partial_\rho \xi^\mu - p_{A_\nu}^\mu \partial_\rho \xi^\rho + p_{A_\rho}^\mu \partial_\rho \xi^\nu\right)\partial_{p_{A_\nu}^\mu}$$

The conserved multisymplectic $(n-1)-$form is:

$$G_\xi = p_{A_\nu}^\mu A_\rho \partial_\nu \xi^\rho d^{n-1}x_\mu + \xi^\alpha p_{A_\nu}^\mu dA_\nu \wedge d^{n-2}x_{\alpha\mu} + \mathcal{H}\xi^\mu d^{n-1}x_\mu$$

and the conserved charge, once integrated by parts, is:

$$\boxed{Q_\xi = \int_{\Sigma_t} \pi^0 A_\mu \partial_0 \xi^\mu d^3x + \int_{\Sigma_t} \xi^\mu\left(\pi^\nu \partial_\mu A_\nu - A_\mu \partial_i \pi^i - \pi^i \partial_i A_\mu - \delta_\mu^0 \mathcal{L}\right) d^3x + \oint_{\partial\Sigma_t} \xi^\mu \pi^i A_\mu d^2x_i}$$

One advantage here of the multisymplectic formalism with respect to the symplectic $(3+1$ split$)$ is that, in the latter, charges cannot be canonically integrated because of the boundary terms[4].

This conserved $(n-1)-$forms fulfil a closed algebra with respect to the graded Poisson brackets:

$$\{F_\varepsilon, F_\zeta\} = 0, \qquad \{F_\varepsilon, G_\xi\} = F_{\mathcal{L}_\xi \varepsilon}, \qquad \{G_\xi, G_\eta\} = G_{\mathcal{L}_\eta \xi}$$

General Relativity

General Relativity is invariant under all diffeomorphisms $x^\mu \to x^\mu - \xi^\mu(x)$. This transformation is generated in the multisymplectic phase space by[3]:

$$V_\xi = -\xi^\mu \partial_\mu + \left(g_{\rho\nu}\partial_\mu \xi^\rho + g_{\mu\rho}\partial_\nu \xi^\rho\right)\partial_{g_{\mu\nu}} - \left(p_{g_{\rho\sigma}}^\nu \partial_\nu \xi^\mu - p_{g_{\rho\sigma}}^\mu \partial_\nu \xi^\nu + p_{g_{\nu\sigma}}^\mu \partial_\nu \xi^\rho + p_{g_{\rho\nu}}^\mu \partial_\nu \xi^\sigma\right)\partial_{p_{g_{\rho\sigma}}^\mu}$$

The conserved $(n-1)-$form is:

$$G_\xi = p_{g_{\rho\sigma}}^\mu \xi^\nu dg_{\rho\sigma} \wedge d^{n-1}x_{\nu\mu} + p_{g_{\rho\sigma}}^\mu\left(g_{\nu\sigma}\partial_\rho \xi^\nu + g_{\rho\nu}\partial_\sigma \xi^\nu\right) d^{n-1}x_\mu + \mathcal{H}\xi^\mu d^{n-1}x_\mu$$

and the conserved charge, once integrated by parts, is:

$$\boxed{Q_\xi = 2\int_{\Sigma_t} \pi_\mu^0 \partial_0 \xi^\mu d^3x + \int_{\Sigma_t} \xi^\mu\left(-2\partial_i \pi_\mu^i + \pi^{\rho\sigma}\partial_\mu g_{\rho\sigma} - \delta_\mu^0 \mathcal{L}\right) d^3x + 2\oint_{\partial\Sigma_t} \xi^\mu \pi_\mu^i d^2x_i}$$

The boundary integral is the ADM momentum and the bulk part also contains total derivatives in $\mathcal{L} = R\sqrt{-g}$ that, given a suitable asymptotic expansion of the parameter, go to boundary terms. When including the lagrangian of the boundary[5] we also recover ADM mass.

The problem of the closed form

Given a vector field X, the equation $\iota_V \Omega = dF$ does not determine **univocally** the generator F because there is a freedom to add to the generator a **closed form** ω such that $d\omega = 0$ while the equation still holds.

This freedom does not affect the charges in the bulk, but may **contribute to their boundary part**, to the point that it can make the boundary contribution to the charge vanish.

For example, we can add to the gauge $(n-1)-$form of electromagnetism the following closed form:

$$F'_\varepsilon = p_{A_\nu}^\mu \partial_\nu \varepsilon d^3x_\mu - d\left(\varepsilon p^{0i} d^2x_{0i}\right)$$

so that, when integrated on the constant-time hypersurface Σ_t, we obtain a pure boundary contribution, which cancels exactly the boundary term of the charge:

$$Q'_\varepsilon = \int_{\Sigma_t} \pi^0 \partial_0 \varepsilon d^3x - \int_{\Sigma_t} \varepsilon \partial_i \pi^i d^3x$$

However, this cannot be done for every submanifold Σ of the spacetime, but **only with one chosen submanifold**. In this way, the $(n-1)-$form F'_ε will still have boundary terms if integrated over any other submanifold different from Σ_t, revealing that the geometric structure in the boundary might still underlie the comparison between different submanifolds. Moreover, electric charge is no more a scalar.

Conclusions and outlook

- Multisymplectic formalism allows for a direct computation of the boundary terms.
- Boundary terms on the conserved charges arise from the derivatives of the parameter after integration by parts.
- In contrast to the symplectic formalism, generators in the multisymplectic formalism are always integrable, regardless of boundary terms.
- Further research on the role that the closed form has in the computation of boundary terms is needed.

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Acknowledgements: I would like to thank I. Kanatchikov and Javier Matulich for their infinitely valuable teachings, and Jordi Gaset for his comments and for providing the funding to print this poster.